MeteoGlobe 🌍

A 3D interactive weather globe. The Earth spins in your browser — click any country or city to see live weather conditions, forecasts, and animated sky effects (rain, snow, sun…). Weather icons come from the official **MeteoSwiss** pictogram set. Zooming into Switzerland reveals live data from ~170 real measurement stations.

Requirements

- Docker Desktop
- A free OpenWeatherMap API key (see below)

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1 — Get your free API key

1. Go to https://openweathermap.org/api
2. Click **Sign Up** and create a free account
3. After signing in, go to **API keys** (top-right menu → your username)
4. Copy the default key that was created for you
5. Wait ~10 minutes before using it — new keys need time to activate

The free plan gives you **1 000 calls/day**, which is more than enough for personal use.

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2 — Setup & run

Paste your key into `docker-compose.yml` (one line to edit):

```
OWM_API_KEY: paste_your_key_here
```

Then start the app:

```
docker compose up -d --build
```

Open http://localhost:3000 in your browser — done.

> The first build downloads Python and installs dependencies (~1 min).
> Every start after that is instant (`docker compose up`).

To stop:

```
docker compose down
```

3 — Uninstall (remove everything)

Stop the app and delete all Docker data it created (image, container, cache):

```
docker compose down --rmi all
```

Then delete this folder. That's it — nothing else is installed on your machine.

Project structure

```
Meteo-app/
│
├── docker-compose.yml     ← only file you need to edit (your API key)
├── Dockerfile             ← builds the Python image
├── server.py              ← FastAPI backend (weather proxy + icon cache)
├── requirements.txt       ← Python dependencies (installed inside Docker)
│
└── public/                ← frontend (HTML + CSS + JS, no build step)
    ├── index.html
    ├── style.css
    ├── app.js             ← globe, markers, search, panel, screensaver
    └── weather-fx.js      ← canvas weather animations (rain, snow, sun…)
```

Troubleshooting

Problem	Fix
Globe is black	Check browser console (F12) — usually Globe.gl CDN unreachable
401 errors in console	OWM key is wrong or not yet active (wait 10 min)
Weather icons show as emoji	MeteoSwiss CDN unavailable — emoji fallback is intentional, data is still correct
Port 3000 already in use	Change both `3000` values in `docker-compose.yml` to e.g. `3001`